

Hazel Lim

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[LinkedIn](#) | [Portfolio](#) | [GitHub](#)

EDUCATION

Columbia University, *Master of Science Degree in Computer Science, expected* Dec 2026
Track in Computer Vision, Graphics, Interaction, and Robotics; GPA: 3.95
Honors: Lustgarten-Whitney Family Fellowship
Relevant Coursework: Databases, Linear Algebra, Data Structures, Deep Learning for CV, Neural Rendering Graphics

Northeastern University, *Bachelor of Science Degree in Psychology and Criminal Justice* May 2024
Minor in Computer Science; GPA: 3.98
Honors: Dean's List, NU Honors Program, Amelia Peabody Scholarship
Relevant Coursework: Cognition, Calculus, Neuropsychology, Statistics, HCI, Object-Oriented Design

RESEARCH EXPERIENCE

Laboratory of Neural Engineering and Control (Graduate Research Assistant) New York, NY
Columbia University | Supervisor: Dr. Qi Wang Feb 2026 – present

- Investigating the coupling between pupil dilation dynamics and EEG patterns using existing physiological datasets, extending prior work on the locus coeruleus–norepinephrine system's role in pupil-linked arousal and cortical state.
- Benchmarking deep learning architectures, including temporal models (TCNs, LSTMs) and attention-based approaches, to predict EEG signals from pupillometric inputs across varying arousal states.
- Motivated by applications in adaptive learning environments, where real-time pupil-EEG coupling could inform BCI systems capable of detecting and responding to fluctuations in focus and cognitive engagement.

Columbia University Irving Medical Center (Graduate Research Assistant) New York, NY
Columbia University | Supervisor: Dr. Yuan Yading Jan 2026 – present

- Developing a NeRF-based reconstruction pipeline for CT scan data, extending neural rendering to 3D volumetric medical imaging.
- Conducting a structured literature review of current approaches to neural radiance fields and Gaussian splatting in medical imaging, synthesizing methodological advances to inform pipeline development.
- Building and validating data preprocessing workflows for high-dimensional scan inputs, with attention to data integrity and domain-specific constraints relevant to safety-critical systems.

Graphics Imaging Light Measurement Lab (Graduate Research Assistant) New York, NY
Columbia University | Supervisor: Dr. Corey Taylor Franklin Sep 2025 – Jan 2026

- Evaluated zero-shot learning pipelines for action and behavior recognition using 3D CNNs (R(2+1)D-18) and vision-language models (CLIP, SigLIP) on UCF-101 and real-world camera trap datasets.
- Assessed model performance across distribution shifts and prompt variations — work directly analogous to benchmarking and evaluation of generative model behavior.
- Adapted CLIP for zero-shot inference on novel datasets; implemented end-to-end evaluation pipelines with GPU runtime profiling.

PROJECTS

Non-Invasive EEG Speech Decoding with Semantic Comprehension Scoring | Neurohackathon, San Francisco

- Built an end-to-end EEG-to-text decoding pipeline using contrastive learning (SigLIP) to align 8-channel EEG brain signals to T5 word embeddings, achieving 66.7% top-10 word retrieval accuracy on a held-out test set.
- Implemented N400 amplitude extraction and theta band power (4–8 Hz) computation per word epoch as neural markers of semantic processing difficulty, integrating them as additional features into the EEG encoder to augment word decoding with comprehension scoring.
- Collected and preprocessed real EEG recordings using a BCI Core-8 headset under an RSVP paradigm, validating N400 semantic violation detection ($t = 1.67$, $p = 0.097$, Cohen's $d = 0.22$) at posterior parietal sites consistent with established neuroscience literature.
- Extended the decoding pipeline with an LLM-based semantic correction module that uses per-word N400 surprise signals to guide sentence reconstruction, targeting adaptive education applications where comprehension difficulty can be detected concept-by-concept from brain activity.

Three-Dimensional Reconstructions of the Brain Using Neural Radiance Fields

- Implemented an MRI neural rendering pipeline using PixelNeRF (conditional NeRF) to synthesize novel viewpoints from limited inputs (single MRI brain scan).
- Curated and preprocessed a multi-site MRI dataset of ~860 slices (UCSF: 461, BraTS 2020: 369, Whole-brain: 30), including RAS reorientation, voxel-size resampling, foreground cropping, intensity normalization, and synthetic 360° slice generation for NeRF training.
- Trained a PixelNeRF-based MRI renderer that predicts voxel-wise physical contrast parameters and performs volumetric rendering to output grayscale intensity per ray.

Estimating Brain Age Gap from Handwriting and Drawing Datasets to Study Neurodegenerative Diseases

- Designed and implemented a deep learning pipeline to estimate behavioral brain age gap from handwriting and drawing images, comparing image-based backbones (ResNet-18, ResNet-50, ViT) and a custom feature-extraction MLP.
- Developed a 63-dimensional stroke-feature MLP with polynomial feature interactions, achieving the highest age-group classification performance in our study (~86% test accuracy), outperforming fine-tuned ResNet-50 on the same data.
- Applied the trained age models to Parkinson's and Autism Spectrum Disorder's handwriting/drawing data to estimate behavioral brain age gap and compare PD/ASD participants with age-matched healthy controls, exploring its potential as a low-cost screening signal complementary to MRI-based biomarkers.

TECHNICAL SKILLS

Languages & Frameworks: Python, PyTorch, HuggingFace Transformers, NumPy, OpenCV, Git, Java, C

Deep Learning: CNNs, Transformers (ViT, CLIP), Contrastive Learning, VAEs, NeRF / Neural Rendering, Transfer Learning, Zero-Shot Learning

Research Methods: Model evaluation & benchmarking, EEG/pupillometry data collection, literature synthesis, experimental design, statistical analysis

ADDITIONAL WORK EXPERIENCE

Focus on Cognitive Impairment Lab

Boston, MA

Research Assistant

Feb 2023 - May 2024

- Conducted and analyzed data from cognitive and EEG tests, providing crucial data for ongoing research on cognitive changes in memory and attention associated with mobile technology
- Gathered and synthesized relevant sources and data for a literature review on cognitive impairment and its intersection with age
- Edited existing protocols to reflect ongoing changes in experimental sessions, maintaining precision and alignment with research goals

Build It Yourself

Cambridge, MA/Hybrid

Workshop Leader and Developer

Jan 2023 - May 2025

- Led after-school workshops and discussions on-site in Cambridge and remotely in Mexico
- Taught students how to plan and develop their own machines and games through Scratch and PowerPoint
- Encouraged students to design characters and lab books and provide feedback when necessary

North Suffolk Mental Health Association

Boston, MA

Case Manager, Clinical Support Specialist

Feb 2022 - Jul 2022

- Facilitated an intensive group treatment program for clients with schizophrenia
- Developed skill building interventions, organized treatment plan documentations and assessments, and coordinated treatment with providers, guardians, and families
- Ensured safety of the residents in community, maintained a nurturing program environment, and assisted with daily living activities